

10-Minute English PFE Defense

Presenter memory sheet · target delivery 9:15 · 45-second safety margin

One-sentence thesis

I built a personalized, context-aware next-POI engine, decoded it without retraining into loop-free itineraries, and evaluated each task on its own fair benchmark.

0.187

HR@1 among ~5,100 NYC candidates

0.289 > 0.259

Frozen Rollout beats Trained Pointer

0.021

maximum Random gap vs Chen 2016

The rule to memorize

Compare results only when the dataset, protocol, and metric are identical.

Never compare next-POI HR@k with itinerary pairs-F1, and never compare NYC itinerary scores directly with Flickr literature scores.

Slide	Time	Memory cue
1	0:25	Smart Visit converts candidates into one coherent route.
2	0:55	A useful trip needs both selection and correct order.
3	1:00	Existing methods trade off personalization and route feasibility.
4	1:00	One pipeline contains two distinct evaluation spaces.
5	1:10	Graph, context, user, and sequence signals produce one ranking.
6	1:05	Frozen Rollout adds route constraints without changing the model.
7	0:45	The engine is competitive, but not state of the art.
8	1:00	Frozen Rollout wins, and dense prefix supervision helps explain why.
9	1:15	Flickr makes the literature comparison valid and honest.
10	0:40	Score, decode, and evaluate honestly.

If running late: use each slide's 'Skip if late' line; if needed, compress slides 5 and 9, but keep the state-of-the-art gap and the comparability rule. Safe closing: *In this tested setting, decoding a densely supervised next-POI engine outperformed a dedicated itinerary model.*

Full 9:15 Talk Track

Pronunciation: say 'HR at one', 'HR at ten', and 'pairs F-one'. Each slide includes one sentence to memorize, a transition, and a safe sentence to skip.

Slide 1 · 0:25 · Smart Visit: From 5,100 Places to One Coherent Itinerary

MEMORY: Smart Visit converts candidates into one coherent route.

Good morning. Smart Visit asks how roughly 5,100 candidate places can become one personalized, coherent tourist itinerary. I will follow five parts: the problem, the state of the art, the solution, its implementation, and the conclusion.

Transition: First, what problem are we solving?

Skip if late: I will follow five parts in this presentation.

Slide 2 · 0:55 · A Good Trip Answers Two Questions

MEMORY: A useful trip needs both selection and correct order.

A tourist system must answer two different questions: what should this user visit next, and in what order should several places form a complete trip? A ranked list is not yet an itinerary; it may repeat locations, ignore the destination, or create a poor order. The research question is whether a personalized, context-aware next-POI engine can be decoded into a loop-free route without confusing next-step accuracy with itinerary quality.

Transition: Now consider how the literature approaches this problem.

Skip if late: A ranked list is not yet an itinerary.

Slide 3 · 1:00 · Existing Methods Trade Off Personalization and Feasibility

MEMORY: Existing methods trade off personalization and route feasibility.

Three broad families dominate the literature. Route-optimization methods such as OP and TTDP enforce constraints by construction, but often use simple utility scores. Personalized and context-aware recommenders learn relevance, yet usually rank individual POIs rather than generate a constrained route. Deep sequential models learn mobility transitions, but next-POI objectives and greedy decoding do not guarantee global coherence. No family fully unifies personalization, context, ordering, and route feasibility. The central gaps are therefore the connection between prediction and itinerary quality, and fragmented evaluation. This thesis addresses those two gaps directly; cold-start and richer environmental context remain partial gaps.

Transition: These gaps determine the proposed solution.

Skip if late: Cold-start and richer environmental context remain partial gaps.

Slide 4 · 1:00 · One Engine, One Decoder, and Two Fair Evaluations

MEMORY: One pipeline contains two distinct evaluation spaces.

The solution deliberately separates two phases. Phase one trains a personalized next-POI engine on Foursquare New York and evaluates it with HR, NDCG, and MRR. Phase two turns its scores into routes through Frozen Rollout, compared internally with a Trained Pointer using set-F1 and pairs-F1. Since New York itinerary values cannot be compared with Flickr literature values, the itinerary task is repeated on the Flickr Benchmark under the Chen-2016 protocol. The two tasks never share one leaderboard.

Transition: Let us now open the prediction engine.

Skip if late: The two tasks never share one leaderboard.

Slide 5 · 1:10 · The Engine Combines Places, Context, and User Preferences

MEMORY: Graph, context, user, and sequence signals produce one ranking.

The implemented engine combines four signals. A hybrid graph links POIs through training-only co-visits and ten geographic neighbours, and a two-layer GCN creates place features. Distance and elapsed-time gaps provide lightweight context. A GRU summarizes the visited sequence, while a learned user embedding represents identity-based preference. The GRU state and user embedding feed an MLP that scores the full vocabulary of about 5,100 POIs. Each session contributes every prefix-to-next-POI pair, producing on the order of one hundred thousand supervised examples. Training uses Adam, dropout, gradient clipping, and early stopping on validation HR at ten.

Transition: The same trained engine is then reused as a route scorer.

Skip if late: Training uses Adam, dropout, clipping, and early stopping.

Slide 6 · 1:05 · Frozen Rollout Turns Scores into a Loop-Free Route

MEMORY: Frozen Rollout adds route constraints without changing the model.

Frozen Rollout creates an itinerary without retraining the engine. A query provides a user, a start POI, an optional destination, and the desired length. At each step, the frozen engine scores all candidates; the decoder masks visited POIs, reserves a supplied destination for the final step, selects the best valid candidate, and repeats. Greedy is the headline decoder, while beam search keeps several partial routes. When a destination is supplied, the method guarantees it as the last stop; in all cases, the route is loop-free and has the requested length. It does not claim a global optimum, opening-hour feasibility, or travel-time optimization.

Transition: We can now evaluate the engine and the generated routes.

Skip if late: Beam search keeps several partial routes.

Slide 7 · 0:45 · Competitive, but Not State of the Art

MEMORY: The engine is competitive, but not state of the art.

On New York, the engine obtains HR at one of 0.187, HR at five of 0.476, HR at ten of 0.588, and MRR of 0.316. HR at one means the exact next POI ranks first among roughly 5,100 candidates almost one time in five. The result is slightly above STGCN at 0.180, but below GETNext at 0.240 and LLM4POI at 0.340. Its honest position is the LSTM/STGCN tier, not the state of the art.

Transition: The itinerary experiment gives the more surprising result.

Skip if late: Its honest position is the LSTM-STGCN tier.

Slide 8 · 1:00 · Frozen Rollout Wins; Dense Supervision Helps Explain Why

MEMORY: Frozen Rollout wins, and dense prefix supervision helps explain why.

On 2,880 New York itineraries of length at least three, Frozen Rollout reaches 0.289 pairs-F1, compared with 0.259 for the Trained Pointer. Beam search reaches only 0.290. The strongest explanation is supervision density: the engine learns from every prefix, on the order of one hundred thousand prefix-target examples, whereas the Trained Pointer sees one complete trajectory per session and trains from scratch. The lesson is not that pointer models never work; it is that a densely supervised scorer is difficult to beat. These itinerary metrics are never compared with next-POI HR.

Transition: But New York is not a valid literature benchmark.

Skip if late: Beam search reaches only 0.290.

Slide 9 · 1:15 · Flickr Validates the Protocol and Shows the Remaining Gap

MEMORY: Flickr makes the literature comparison valid and honest.

For fair literature comparison, the Flickr Benchmark follows the Chen-2016 protocol: leave one trajectory out, provide the first and last POI plus route length, and evaluate loop-free routes of length at least three with pairs-F1. Our Random results reproduce Chen's published Random baseline within a maximum absolute gap of 0.021, validating the harness. On that published scale, our classical baselines span 0.23 to 0.59, and the Trained Pointer spans 0.31 to 0.49, while SelfTrip and AR-Trip remain around 0.80. Personalization is conditional: the user embedding changes pairs-F1 by plus 0.038 in Glasgow, minus 0.007 in Osaka, and minus 0.017 in Toronto. These deltas come from a separate user-embedding ablation, not the main pointer run. Sparse leave-one-out history therefore exposes cold-start. Other limits are New-York-only engine training, offline evaluation, context restricted to distance and elapsed time, and no opening-hour or scheduling constraints. New York and Flickr scores are never compared directly.

Transition: I will close with the three results to retain.

Skip if late: New York and Flickr scores are never compared directly.

Slide 10 · 0:40 · Three Results to Remember

MEMORY: Score, decode, and evaluate honestly.

Three results matter. The engine reaches HR at one of 0.187. Frozen Rollout converts its scores into loop-free itineraries and reaches 0.289 pairs-F1, above the Trained Pointer at 0.259 on the same New York protocol. The Flickr reproduction validates the literature comparison. The contribution is therefore a reproducible foundation: a personalized scorer, a simple effective decoder, and an honest evaluation. Thank you.

Transition: *Thank you; I welcome your questions.*

Skip if late: *The contribution is a reproducible foundation for future work.*

Likely Jury Questions

Answer directly, give one piece of evidence, and stop. Return to the closing slide after the answer.

1. What is the main contribution of this thesis?

A reproducible context-aware next-POI engine, Frozen Rollout for converting its scores into loop-free routes without retraining, and a Flickr Benchmark that places the itinerary results on a literature-comparable scale. The contribution is the system, the integration experiment, and the rigorous evaluation - not a new state-of-the-art architecture.

2. Why treat next-POI prediction and itinerary generation as separate tasks?

Next-POI prediction ranks one future place, while itinerary generation must select and order an entire route. They therefore use different outputs and metrics: HR@k, NDCG, and MRR for next-POI; set-F1 and pairs-F1 for itineraries.

3. What does HR@1 = 0.187 actually mean, and why use a GRU?

It means the correct next venue is the top prediction in 18.7 percent of cases among about 5,100 candidates; it is not an itinerary-success rate. The GRU provides a lean, reproducible sequential backbone and produces an honest LSTM/STGCN-tier result.

4. How does Frozen Rollout generate an itinerary?

It repeatedly applies the frozen next-POI scorer to the current partial route. Already visited POIs are masked, the requested end POI is reserved for the final step, and greedy or beam selection continues until the requested route length is reached.

5. Why did Frozen Rollout outperform the Trained Pointer in this setup?

The strongest explanation is supervision density: the engine learns from roughly 10^5 prefix-target pairs, whereas the Trained Pointer receives one whole-trajectory example per session. Frozen Rollout reuses a converged engine, while the Trained Pointer trains its encoder from scratch. Adding context improves the Trained Pointer only from 0.259 to 0.261.

6. How do you know the Trained Pointer result is not a training bug?

Validation pairs-F1 rose to a genuine peak before early stopping, and the context-enhanced variant was tested. The same data-size pattern appears on the Flickr Benchmark, where the Trained Pointer trails Markov. This supports a real negative result for the tested setup, not a universal claim about every itinerary model.

7. Why does beam search barely improve Frozen Rollout?

Beam search changes pairs-F1 only from 0.289 to 0.290. The much higher set-F1 of 0.609 shows that many correct places are found but their order remains imperfect. Searching harder over the same local scores cannot introduce information the scorer never learned.

8. Why can the NYC and Flickr results not be compared directly?

NYC contains about 5,100 POIs, while the Flickr cities contain only 27 to 88. Chance level and task difficulty differ dramatically. Valid comparison requires the same dataset, protocol, and metric.

9. How did you validate the Flickr evaluation harness?

I followed Chen 2016: leave-one-trajectory-out evaluation, first and last POIs plus route length given, routes of length at least three, loop-free output, and pairs-F1. Our Random results differ from Chen by at most 0.021 across five cities.

10. Does personalization actually help?

Conditionally. The user embedding improves Glasgow from 0.451 to 0.489, a +0.038 gain, but changes Osaka by -0.007 and Toronto by -0.017. Under leave-one-out, many users have too little remaining history to estimate an identity embedding reliably.

11. What are the main limitations?

The engine remains below neural and LLM state of the art; decoding cannot correct the scorer's myopia; context is limited to distance and elapsed-time gaps; decode-time elapsed time is assumed; and the next-POI experiment covers one city with offline evaluation only - no deployment or user study.

12. What is future work, and what was not implemented?

Future work includes weather and time-of-day context, persona or content-based cold-start representations, schedule-aware decoding with travel cost and opening hours, stronger pre-training, multi-city validation, and user studies. The orienteering or iterated-local-search solver was scoped but not implemented.

Never claim: state of the art; that HR@1 and pairs-F1 are comparable; that NYC and Flickr have the same difficulty; that personalization always helps; or that the orienteering / ILS method was implemented.